Derivation appendix: 08 — the radiative decays of the tau in the soft-broken vacuum

tau to mu gamma and tau to e gamma

1. $\ell_i \to \ell_j \gamma$ amplitudes for general (i, j)

derived here — §1

Claim

$$c_L = \sum_f \frac{\frac{1}{2} I_1(t_f)(m_j Y_{jf} Y_{if}^* + m_i Y_{fj}^* Y_{fi}) + I_2(t_f) m_f Y_{fj}^* Y_{if}^*}{4 m_i m_\phi^2}$$

Independent checks

- (μ, e) reproduces notebook 07's stored results at 25 points
- flavour relabelling maps (τ, ℓ) onto the (μ, e) code exactly

HKZ13 (A3) with external masses kept, for $\tau \to \mu\gamma$, at 700 and 25 GeV

OST16 (19), the explicit $\tau \to \mu\gamma$ one loop

Would be falsified by: an m_τ/m_μ flip factor carried over from $\mu \to e\gamma$ (falsify run)

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2. $\mathcal{B}(\tau \to \ell\gamma) = 48\pi^3 \alpha(|A_L|^2 + |A_R|^2)/G_F^2 \times \mathcal{B}(\tau \to \ell\nu\bar{\nu})$

imported — §2 ([OST16] (17))

Independent checks

- equals the [HKZ13] lifetime normalization to below 1 percent
- reproduces both [HKZ13] τ bounds with their W number

Would be falsified by: the two-loop pieces with the wrong relative sign (falsify run)

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3. Residual $\mathbb{Z}_2$: $\tau \to e\gamma = 0$, $\tau \to \mu\gamma \neq 0$ at the exact vacuum

derived here — §3–4

Independent checks

- $\mathcal{B}(\tau \to e\gamma) < 10^{-24}$ at 1973 exact points $\times$ 60 values of μ_3
- 3 odd states without $\tau\mu$, 4 even states without e couplings, at every evaluation
- $\mathcal{B}(\tau \to e\gamma) \propto (60^\circ - \varphi)^2$ off the alignment
- all three channels identical at the S_3 image (20 points)

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4. Quark couplings along the vacuum direction

assumed — §5

Independent checks

- same ansatz as notebooks 06 and 07; three variants bound its effect

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